

Problem: [Target persona: elementary, lower] A baker bakes 24 muffins and shares
them equally among 6 friends. How many muffins does each friend get?

SFT (Qwen3-1.7B)

Let x be the number of muffins each friend receives.
Set up the linear equation $6x = 24$, where x is the
unknown variable.

Solve the equation by dividing both sides by 6: $x =$
 $24 / 6 = 4$.

Therefore, the solution to the equation is $x = 4$.

(Uses algebra/equation/variable — inappropriate for
an elementary learner.) ✗

+ BC-StepDPO (ours)

We have 24 muffins to share fairly with 6 friends.

Let's split the 24 muffins into 6 equal groups.

Sharing 24 into 6 equal groups: $24 \div 6 = 4$.

So each friend gets 4 muffins.

(Plain, friendly language with no algebra — fits the
persona.) ✓

Figure 3. SFT vs. BC-StepDPO: matching the target persona's register (elementary student).